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Well-being and organizational performance: An organizational-level test of the happy-productive worker hypothesis

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It is often assumed that happy workers are also productive workers. Although this reasoning has frequently been supported at the individual level, it is still unclear what these findings imply for organizational performance. Controlling for relevant work characteristics, this study presents a large-scale organizational-level test of the happy-productive worker hypothesis, assuming that high individual well-being leads to high individual-level performance, which should translate into high organizational performance (such as high efficiency and productivity). Job-specific employee well-being was measured as job satisfaction and emotional exhaustion. Using data from 66 Dutch home care organizations, the relationships among aggregated levels of demands, control, support, emotional exhaustion and satisfaction on the one hand, and organizational performance on the other, were examined using regression analysis. The hypotheses were partly confirmed, especially high aggregated levels of emotional exhaustion were related to low organizational performance. Although these findings support the reasoning that happy organizations are indeed productive organizations, more theorizing and more longitudinal research on the associations between individual-level well-being and organizational performance is imperative to improve understanding of these relationships. The findings underline the importance of improving worker well-being: this is not only important for individual workers, but may also have positive consequences for organizations and their clients.

Keywords: emotional exhaustion; satisfaction; organizational performance; happy-productive worker hypothesis; well-being work-related stress

Introduction

One often-voiced idea in occupational health psychology is that satisfied and happy workers are on average more productive than others. This *happy-productive worker hypothesis* dates from the Human Relations Movement of the 1930s (Wright, Cropanzano, & Bonett, 2007), but is still popular today. Interestingly, the evidence for this hypothesis is mixed. For example, Vroom (1964) reviewed 20 individual and group-level studies on the relationship between workers' job satisfaction and productivity, finding that the median correlation between these concepts was only .14. Similarly, based on a meta-analysis of 74 studies examining the association between performance and both general and facet-specific (e.g., satisfaction with pay or promotion opportunities) measures of job satisfaction, Iaffaldano and Muchinsky

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(1985) obtained a mean correlation of .17, concluding that satisfaction and performance are “only slightly related” (p. 269). Integrating findings from 20 studies, Petty, McGee, and Cavender (1984) were more optimistic: they reported a mean correlation of .31 between overall job satisfaction and performance, concluding that “the size of the mean correlation [...] between overall satisfaction and job performance is *impressive*” (p. 719; our italics). In their meta-analysis of 312 studies, Judge, Thoresen, Bono, and Patton (2001) obtained a .30 correlation between employee job satisfaction and employee productivity. Finally, Taris’ (2006a) meta-analysis of five studies on the relationship between emotional exhaustion (the core component of the burnout syndrome) and objectively recorded in-role performance revealed a corrected correlation of $-.22$.

On the one hand, these findings support the basic tenet of the happy-productive worker hypothesis; i.e., contented, satisfied workers perform on average better than other workers. On the other hand, it is clear that the magnitude of this association should not be overestimated. Indeed, although previous research usually shows that “happy” workers are somewhat more productive than others, it is still unclear what these findings imply at the organizational level: do organizations with dissatisfied and exhausted workers on average perform *noticeably* worse than other organizations (cf. Judge et al., 2001; Ostroff, 1992)? The answer to this practically important question is not immediately clear, not only because the individual-level relationships between worker satisfaction and exhaustion on the one hand and productivity on the other are relatively weak, but also because previous research has often suffered from a number of limitations. Firstly, the fact that the association between employee well-being and productivity is weak brings in the possibility that third variables are responsible for this relationship (Judge et al., 2001; Wright et al., 2007). For example, job satisfaction and emotional exhaustion are associated with work characteristics such as job control and job demands (De Lange, Taris, Kompier, Houtman, & Bongers, 2003), and these concepts may also be related to performance. Thus, the association between measures of worker well-being and productivity might be due to uncontrolled third variables.

A second limitation of previous research concerns the measurement of performance. Performance can be measured in a variety of ways. For example, Roe (1999) distinguished between performance measures that refer to the congruence between the goal to be reached and the outcomes of the efforts of reaching that goal, versus the quality of the process that leads towards attaining a goal. The first type of measure focuses on objective, quantitative measures of performance (e.g., profit and sales), whereas the second type concentrates on process-oriented measures such as customer and patient satisfaction. Whereas it is likely that quantitative and process-related measures of performance are related (every result is at least partly determined by the process that generated it), conceptually it is important to acknowledge that “performance” may refer to very different concepts, and that the association between worker well-being and performance may vary with the type of outcome considered. For example, Quinn (1988) suggests that high levels of worker well-being may coincide with *low* performance (in terms of low efficiency and productivity), whereas Loveman (1998) proposes that high worker well-being is associated with *high* performance (e.g., customer loyalty and satisfaction). Unfortunately, the distinction between different types of performance indicator has as yet received little attention.

The present study addresses these issues in an organizational-level study into the associations between measures of worker well-being (that is, average levels of job satisfaction and emotional exhaustion) and various indicators of organizational performance. For the purpose of this study, information was collected on the performance of 66 organizations in the Dutch domiciliary care sector. The employees of these organizations were surveyed about both their well-being and their job. Using these data, the present study tests the happy-productive worker hypothesis (a) at the level of the participating organizations; (b) controlling for relevant work characteristics; (c) using two indicators of well-being; and (d) using multiple measures of organizational performance. The questions addressed are (1) is there a relationship between the average level of employee well-being in organizations and organizational performance, and (2) does the strength of this association vary across different types of well-being and performance?

Worker well-being and performance: individual-level approaches and findings

Organizational-level performance and individual-level well-being are theoretically often connected with each other by assuming that individual-level well-being leads to higher levels of individual-level performance (Judge et al., 2001; Taris, 2006a), in turn leading to better organizational performance (Bakker, Van Emmerik, & Van Riet, 2008; Koys, 2001). Therefore, below we first discuss the relationships between individual-level well-being and individual-level performance.

Underlying processes. Subjective well-being is often construed as a global, affective state (Diener, Suh, Lucas, & Smith, 1999). In contrast, researchers in work and organizational psychology have found it useful to focus on measures of job-specific well-being, i.e., as it is experienced in the work context (Van Horn, Taris, Schaufeli, & Schreurs, 2004; Warr, 2007). Previous research has examined the happy-productive worker hypothesis using job satisfaction and (to a lesser degree) emotional exhaustion as indicators of worker happiness. As indicated above, job satisfaction is moderately positively related to employee performance (Judge et al., 2001), whereas emotional exhaustion is negatively related to performance (Taris, 2006a). Based on these findings, it is plausible that both indicators of job-specific well-being are associated with organizational performance. However, the processes underlying these associations may differ.

Satisfaction and performance. The association between job satisfaction and individual performance is frequently accounted for by assuming that satisfied employees feel obliged to do their employer a favour by working hard. This idea stems from Adams's (1963) equity theory, proposing that people both "invest" in their relationships with other actors and receive benefits from that relationship. Equity occurs when these investments are equal to the benefits received. Inequity resulting from the situation in which one invests more (or less) than one receives can be resolved by investing less (or more) in that relationship (Taris, Van Horn, Schaufeli, & Schreurs, 2004).

In the context of the work environment, workers invest in the relationship with their employers in terms of time, effort and knowledge. In return, they receive rewards such as salary, job security, status and prestige. Employees who are satisfied with these aspects would then feel obliged to invest much in the organization in terms of their effort, thus increasing product quality and quantity. This reasoning implies

that primarily affective phenomena such as employee satisfaction also have activating and motivational potential (cf. Bjornebekk, 2008); high (or low) levels of satisfaction (or, more generally, well-being) would trigger high (or low) levels of motivation to perform well.

In line with this reasoning, previous research on the relationship between employee job satisfaction and client satisfaction confirmed that both concepts are positively associated (e.g., Dormann & Kaiser, 2002; Gelade & Young, 2005, for reviews). Heskett, Sasser, and Schlesinger (2002) examine this relationship in terms of value profit chain theory. Again, this approach considers job satisfaction as a motivational force, in that high levels of individual-level satisfaction will lead to higher employee loyalty and increased effort, thus leading to higher individual productivity and better product quality. This results in satisfied and loyal clients, in turn leading to higher profits for the organization.

Emotional exhaustion and performance. Theoretically, a link between emotional exhaustion and individual-level performance may be argued for by proposing that a high level of emotional fatigue (as a milder form of exhaustion, cf. Kristensen, Borritz, Villadsen, & Christensen, 2005) has both an energetic and a motivational component; high levels of emotional fatigue result in being *unable* as well as being *unwilling* to perform well (cf. Thorndike, 1914). Lack of motivation to perform well may be a functional reaction to work overload, and may act as a safety valve that prevents workers from depleting their energetic resources (Schaufeli & Taris, 2005). High levels of exhaustion should therefore lead to low levels of effort expenditure and performance, in turn leading to lower organizational performance. These notions are confirmed by individual-level research showing that high levels of burnout (of which emotional exhaustion is a core dimension) impede work performance (Shirom, Nirel, & Vinokur, 2006; Taris, 2006a), and by Halbesleben and Bowler (2007), who found that high levels of emotional exhaustion were longitudinally associated with low in-role performance; this relation was mediated by the will to achieve the tasks at work.

Individual-level performance versus organizational performance

As indicated above, higher levels of individual-level well-being are often expected to promote higher levels of individual-level motivation and performance, which would lead to higher levels of organizational performance (Bakker et al., 2008; Koys, 2001; Taris, 2006a). At the level of the individual workers, it seems reasonable to assume that well-being and performance are indeed related, even if this association is relatively weak (Iaffaldano & Muchinsky, 1985; Judge et al., 2001; Petty et al., 1984; Taris, 2006a). However, although it appears plausible that high performance of individual workers will positively affect organizational performance, taking this reasoning for granted would be tantamount to making the ecological error of assuming that findings that hold for one level of analysis also apply to another.

In the present case, the idea that there is a one-to-one relationship between high individual-level and high organizational performance is overly simplistic. For instance, an orchestra consisting of players who are all good performers is not necessarily a top-notch ensemble (Klein & Kozlowski, 2000). In order to do well as an orchestra, not only should its individual members perform optimally, but

members of the orchestra should also work together and synchronize their efforts. Thus, high individual performance does *not automatically* lead to high performance as an orchestra; organizational performance is not simply the sum of all individual performances, and may therefore be influenced by factors other than those affecting individual-level performance (Ryan, Schmit, & Johnson, 1996).

Relevant to the present study, this example suggests that high individual performance is not a sufficient condition for high organizational performance. Yet, it seems at least likely that high organizational performance is facilitated by high individual performance, even if other factors may also be relevant in this process (e.g., quality of leadership, Ryan et al., 1996; shared values, Schneider, White, & Paul, 1998). For example, the management literature has provided accumulating evidence that human resource outcomes (including individual-level performance) and organizational performance are related, suggesting that “the proper use of people enhances organizational effectiveness” (Koys, 2001, p. 102). Thus, we assume that individual and organizational performance will be related.

The present study

In order to examine the ideas discussed above, a large-scale survey among 66 Dutch home care organizations and their clients was conducted. Our study included four performance indicators. *Client satisfaction* reflects the degree to which the clients of the home care organizations are satisfied with its services and employees. This not only concerns the quantity of the service delivered, but especially how these are delivered. Thus, this is a process measure of performance (Roe, 1999). Three other indicators primarily tap quantitative performance. *Organizational productivity* refers to the degree to which organizations realize their primary goal, i.e., the production of service hours. The *personnel costs* per paid hour is a measure of the costs made for realizing the output. High personnel costs suggest that an organization's product is too expensive, e.g., due to high overhead costs or personnel that are used inefficiently. *Organizational efficiency* concerns the costs of the product of the home care organizations (the number of hours of care delivered to their clients). Low efficiency implies that the organization fails to employ its resources optimally.

Based on previous theory and findings, we expect organizations with highly satisfied employees (as evidenced by high levels of average employee satisfaction) to be more productive than other organizations, as satisfied employees will expend more effort and work harder (Hypothesis 1a). Moreover, these organizations will be evaluated positively by their clients, as satisfied employees will be motivated and work hard, resulting in higher product quality (Hypothesis 1b). The personnel costs of these organizations will be low, as satisfied employees will work harder and need less time to complete their tasks than employees of other organizations (Hypothesis 1c). Finally, we hypothesize that organizations with satisfied employees will operate more efficiently than organizations with dissatisfied employees, as satisfied employees will work harder and deliver a higher-quality product than dissatisfied employees (Hypothesis 1d).

Our hypotheses for the relationship between emotional exhaustion and organizational performance are the reverse of those formulated for the relationships between job satisfaction and performance. Organizations with exhausted employees (as evidenced by high average levels of emotional exhaustion) should be less productive

than other organizations (Hypothesis 2a); these organizations will be evaluated relatively negatively by their clients (Hypothesis 2b); the personnel costs for these organizations will be relatively high (Hypothesis 2c); and organizations with exhausted employees will work relatively inefficiently (Hypothesis 2d).

Work characteristics and well-being. Finally, our study includes three work characteristics: job demands, job control and social support. According to Karasek and Theorell (1990), these three characteristics cover central aspects of the work situation. Our model proposes that the combination of high levels of job demands and low levels of control and support leads to low worker well-being. For example, reviews of the literature have shown that workers with high demands, low control and low support are indeed more often dissatisfied and exhausted than their colleagues (e.g., De Lange et al., 2003). These reviews have shown that the main effects of demands, control and support on work outcomes are often supported, but that statistical interactions among these factors are largely absent (cf. Taris, 2006b). As Karasek (1998) argued that the form of this interaction is irrelevant, the present study did not test for statistical interactions of demands, control and support.

In the present study, individual-level perceptions of demands, control and support were aggregated to form organizational-level measures of these three job characteristics. We conceptualize the processes linking these concepts and organizational-level performance as isomorphic and functionally equivalent to those operating at the individual level. That is, the constructs at the organizational level are analogous, and the theoretical mechanisms linking them are similar in nature to those at the individual level (DeShon, Kozlowski, Schmidt, Milner, & Wiechmann, 2004). The purpose of the present research is thus to examine whether the individual-level processes for the relationships among job characteristics, well-being (satisfaction and emotional exhaustion) and performance discussed earlier have the expected organizational-level consequences.

Based on these notions, we expect a positive main effect of the average level of job demands in the organization on the average level of emotional exhaustion in that organization (Hypothesis 3a) and a negative effect on the average level of job satisfaction (Hypothesis 3b). Further, we expect a negative effect of the average level of support on emotional exhaustion (Hypothesis 4a) and a positive effect on satisfaction (Hypothesis 4b). Finally, there should be a negative effect of the average level of job control on the average level of emotional exhaustion (Hypothesis 5a) and a positive effect on the average level of satisfaction in that organization (Hypothesis 5b).

Method

Procedure and participants

The data were collected in a study on the quality of work in the Dutch home care sector. Eighty-one of the 95 members of the association of employers in this sector participated in the study (85.3% response). Organizations in the home care sector offer a range of home-support services to their clients in varying circumstances (e.g., clients are pregnant or have just given birth, or are physically or mentally incapacitated). Most employees in this sector are female, and their level of education varies from primary school only (for lower-level jobs) to a college or university

degree (for management jobs). Their tasks include the provision of instrumental, emotional and informational support to their clients, in order to improve their clients' daily functioning (e.g., helping their clients get out of bed and to the toilet, taking care of household chores, and assisting them in developing skills to live more independently). Their services thus help people to keep living in their homes as long as possible.

The present research involved three separate studies; (a) a study among the employees of these organizations; (b) a survey among the organizations' clients, examining their satisfaction with different aspects of the care delivered; and (c) a study on the financial state of affairs of the participating organizations. The employee and the client survey were conducted in the early autumn of 2004. The survey on the performance of the organizations was based on their activities during 2004, as available in 2005.

Employee survey. All employees of all 81 participating home care organizations received a written questionnaire and a postage-paid return envelope at their home addresses, to be returned directly to the researchers. This questionnaire focused on the employees' experienced quality of work. The overall response rate was 48.7% ($N=56,963$). The response rates within the participating organizations varied considerably, from 25% to 68% (the median response rate was 49%). Follow-up analyses revealed that there were no statistically significant differences between the final sample and the population of the 81 organizations in terms of age ($M=42.9$ years, $SD=8.71$) and gender (96% female).

Client survey. Simultaneously with the employee survey, a survey was conducted among 106,637 people who currently used or had used the services of the participating home care organizations during the 2 months preceding the employee survey. These clients received a written questionnaire at their homes, tapping their satisfaction with the services provided, the organization and the worker delivering these services. The questionnaire could be returned to the researchers in a postage-paid return envelope. The response rate was 51.5% ($N=54,987$), which was satisfactory given the length of the questionnaire and the high participant age ($M_{\text{age}}=74.6$ years) (cf. Kaldenberg, Koenig, & Becker, 1994).

Financial survey. The organizations could voluntarily participate in an additional survey on their financial position. This part of the study was conducted by the accountants of a major internationally operating consulting and accounting firm. The present study employs part of the findings of this organization, especially as regards their data on productivity, personnel costs and efficiency. The data pertained to 2004, i.e., the year in which the client and employee surveys were conducted.

Data aggregation and combination of the data. All relevant data resulting from these three studies were aggregated on the level of the participating organizations. Listwise deletion of missing values resulted in a sample of 66 organizations; thus, the present study included 69.5% of the 95 organizations that were members of the association of employers in this sector. The initial number of participating organizations ($N=81$) and the final sample size differed because many organizations did not participate in the financial survey, where participation was voluntary. Together, the 66 organizations participating in this study were responsible for 58.3 billion service hours (i.e., hours during which clients actually received care); this figure is 75% of the total

number of service hours produced by all 81 organizations. Follow-up analysis revealed that the organizations who participated in the financial survey did not differ from those who did not participate in terms of employee satisfaction, emotional exhaustion, job control, job demands and job control, $F(5,72) < 1.00$, *ns*, $R^2 = .03$.

Measures

Employee survey. During the employee survey (which was conducted in the early autumn of 2004), information was collected on many issues. *Job demands* were measured with five items devised by Karasek (1985), including “do you have to work very fast” (1=“never,” 5=“always”). To examine whether aggregation of this concept to form an organizational level of demands was warranted, intraclass correlations (ICCs) were computed (Bliese, 2000). ICC(1) represents the amount of individual-level variance in a particular criterion concept that is explained by the higher-order factor (e.g., the organization). ICC(1) thus varies between 0 (no variance explained) and 1 (100% variance explained). Aggregation is warranted when the *F*-value corresponding with this factor is statistically significant (Klein & Kozlowski, 2000). ICC(2) may be interpreted as the reliability of the group means on a particular criterion variable. It varies between 0 and 1, and is acceptable when it is equal to or exceeds .70. Table 1 shows that both ICCs were acceptable.

Job control was measured using Karasek et al.’s (1998) three-item scale, including “can you decide for yourself how you perform your work?” Inspection of the relevant ICCs revealed that aggregation was warranted. *Social support* was measured using the four-item scale developed by Van Veldhoven and Meijman (1994), including “if necessary, can you ask your colleagues for help?” Table 1 shows that the intraclass correlations were acceptable. *Emotional exhaustion* was measured using Schaufeli and Van Dierendonck’s (2000) adaptation of Maslach, Jackson, and Leiter’s (1996) Emotional exhaustion scale (5 items, including “I feel burned out from my work,” 0=“never,” 6=“every day”). Finally, *employee job satisfaction* was measured with a single self-developed item, namely “Please indicate how satisfied you are with your current job” (1=“extremely dissatisfied,” 10=“extremely satisfied”). Previous research suggested that single-item measures can be acceptable indicators of satisfaction (Wanous, Reichers, & Hudy, 1997). By using a 10-point format we aimed to increase the systematic variance in this concept (as compared to the variance resulting from using more common 4- or 5-point Likert-type scales). Aggregation of the scores on this concept to the organizational level was warranted, see Table 1.

Client survey. The client survey was conducted during the same period as the employee survey (i.e., summer/fall of 2004). The clients who participated in this part of the study received a written 80-item questionnaire, asking them to evaluate 14 aspects of the services provided by the organization and its employees. Again, these aspects were rated on a 10-point report mark format, with the answering categories ranging from 1 (very negative evaluation) to 10 (very positive evaluation). Aspects that were rated included the reliability and flexibility of the organization, and the expertise and

Table 1. Means, standard deviations, reliabilities and correlations among the study variables ($N=66$ organizations).

<i>Variables</i>	ICC(1)	ICC(2)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
(1) Efficiency	–	–	<i>na</i>								
(2) Client satisfaction	.03	.90	.07	.92							
(3) Personnel costs	–	–	–.04	–.15	<i>na</i>						
(4) Productivity	–	–	.33	–.02	–.39	<i>na</i>					
(5) Employee satisfaction	.01	.78	.07	.49	–.22	.00	.78				
(6) Emotional exhaustion	.01	.83	–.11	–.55	.38	–.42	–.54	.85			
(7) Job control	.01	.84	–.06	–.02	–.07	–.03	.36	.05	.61		
(8) Social support	.03	.91	–.04	.40	–.01	–.38	.56	–.08	.22	.71	
(9) Job demands	.02	.90	–.09	–.30	.35	–.42	–.29	.60	–.10	.23	.84
<i>M</i>			91.35	8.26	15.74	63.86	7.53	2.89	6.47	7.26	4.54
<i>SD</i>			6.16	0.19	1.44	3.81	0.21	0.16	0.16	0.42	0.27

Note: Correlations of .23 and higher are significant at $p < .05$ (two-tailed). All values for ICC(1) are significant at $p < .001$ (two-tailed). ICC=intraclass correlation. *na*=not applicable. If applicable, individual-level reliability estimates (Cronbach's alpha) are presented on the diagonal.

friendliness of the employees. The intraclass correlations for the overall scale were acceptable (Table 1).

Financial survey. The financial survey was conducted in 2005 and referred to the organizations' activities during the whole calendar year of 2004. *Organizational productivity* was measured as the number of service hours delivered, expressed as a percentage of the total number of paid hours produced by the employees of the organization. Differences between these two figures are due to, for instance, overhead costs and idle hours. The average productivity in the present research was 63.9% hours ($sd = 3.8$), indicating that the average participating organization spent about two-thirds of its total number of paid hours to providing services to participants.

The *personnel costs* per hour were calculated as the average salary paid by the organization, as specified in the labour contracts of its employees. This is the ratio between the total salary costs and the total number of hours that employees should work according to their contracts. This figure was increased by surcharges for overhead costs and idle hours, resulting in an estimate of the average costs of one service hour.

Finally, *organizational efficiency* was computed using the data envelopment analysis (DEA) method (Seiford & Thrall, 1990). For all participating organizations the costs of one service hour were computed: the organization delivering the cheapest service hours (the *benchmark* organization) was considered 100% efficient. The scores of the other organizations were related to the score of the benchmark. Thus, an organization that produced 90% of the number of service hours produced by the benchmark for the same budget was considered as 90% efficient, and so forth. The average efficiency score of the participating organizations was 91.4% ($sd = 6.16$).

Statistical analysis

Table 1 presents the organizational-level correlations, means and standard deviations for the study variables. The estimates presented in this table are partly based on aggregated individual-level data, implying that information about the individual-level variation in the underlying concepts is lost. This loss of information could bias parameter estimates. However, this bias decreases with increasing sample size (Croon & van Veldhoven, 2007). As the aggregated estimates for the participating organizations were based on relatively large employee samples (average N was 833 participants), the bias resulting from this aggregation procedure will be small (i.e., parameter estimates will be stable across different subsamples).

The data were analyzed using ordinary least squares regression analysis. Preliminary inspection of the data revealed that the data were normally distributed. For the four performance indicators, two blocks of predictor variables were construed. The first block included the two indicators of employee well-being (i.e., satisfaction and emotional exhaustion). The second block contained the three job characteristics (demands, control and social support). Two additional regression analyses were conducted for the two measures of employee well-being, estimating the effects of job characteristics only.

Results

Table 2 presents the standardized effect estimates (betas) for our six series of regression analyses. Focusing on the four performance indicators, the average levels of job demands, job control and social support did not account for a significant additional proportion of the variance in any of the four performance indicators, as shown by ΔR^2 -values that were not statistically significant. Conversely, the block with the two measures of employee well-being contributed significantly to the variance in the performance measures in three out of four cases: only efficiency was unrelated to employee well-being.

Table 2 shows that the average level of employee satisfaction was positively related to client satisfaction (a standardized effect of .29, $p < .05$) but (contrary to our expectations) negatively to productivity (an effect of $-.31$, $p < .05$). Note that the underlying correlation for the latter effect was .00, *ns* (cf. Table 1), meaning that this unexpected effect may be a statistical artifact. Thus, organizations with employees reporting on average high levels of satisfaction also had relatively satisfied clients (Hypothesis 1b supported), but in other respects (efficiency, personnel costs and productivity) they did not perform better than other organizations (Hypotheses 1a, 1c-d not supported).

The average level of employee emotional exhaustion was systematically related to three performance indicators. Standardized effects of $-.36$ on client satisfaction, of .36 on personnel costs, and of $-.58$ on productivity (Hypotheses 2a-c supported, 2d rejected). Thus, organizations in which the average employee reported high levels of emotional exhaustion were evaluated less positively by their clients, their personnel costs were relatively high, and their productivity was lower.

As regards the relationships between the average levels of the job characteristics on the one hand and the aggregated levels of emotional exhaustion and employee satisfaction on the other, we found that the average level of emotional exhaustion in an organization was positively related to the average level of demands and control reported in that organization (effects of .68 and .19, respectively) and negatively to the average level of support ($-.28$) (Hypotheses 3a, 4a, 5a supported). Further, the average level of satisfaction was negatively associated with the average level of job demands in the organization ($-.43$) and positively with average levels of support (.65) and job control (.17) (Hypotheses 3b, 4b, 5b supported).

Discussion and conclusions

This study examined whether organizations whose employees report high levels of well-being (job satisfaction and emotional exhaustion) are more productive than other organizations, controlling for the average levels of job demands, control and support in these organizations. Our findings regarding the associations among organizational-level work characteristics and worker well-being largely replicated individual-level findings for these variables: high average demands, low average support and control were associated with low average well-being (see De Lange et al., 2003, for corresponding individual-level findings). This suggests that at the organizational level, a functionally equivalent process influences average levels of exhaustion and satisfaction as a function of average levels of demands, control and support (cf. DeShon et al., 2004).

Table 2. Results of a series of six ordinary least squares regression analyses for indicators of performance and well-being ($N=66$). Effects are standardized betas).

	A: Performance								B: Well-being	
	Efficiency		Client satisfaction		Personnel costs		Productivity		Emotional exhaustion	Employee satisfaction
	M1	M2	M1	M2	M1		M1	M2		
Employee satisfaction	.01	.10	.29*	.15	-.03	.07	-.31*	-.15		
Emotional exhaustion	-.11	-.01	-.36**	-.30	.36*	.29	-.58***	-.40*		
Job control		-.10		-.16		-.07		.11	.19*	.17*
Social support		-.15		.28		-.06		-.30	-.28*	.65***
Job demands		.00		-.18		.19		-.16	.68**	-.43***
ΔR^2		.02		.06		.03		.09		
R^2	.01	.03	.32***	.37***	.14**	.17**	.25***	.34***	.44***	.57***

Note: M1=model with only measures of employee well-being as predictors, M2=model with job characteristics added as predictors. * $p < .05$; ** $p < .01$; *** $p < .001$.

Regarding the happy-productive worker hypothesis, we found that in a number of cases organizational performance was related to the average level of employee well-being in these organizations. Average levels of the traditional indicator of worker "happiness" (job satisfaction as a measure of job-related affect, cf. Judge et al., 2001; Warr, 2007) were positively related to client satisfaction, but not (or even negatively) to productivity; the two other measures of organizational performance (efficiency and personnel costs) were unrelated to the average level of employee satisfaction in the organizations. Conversely, high average levels of emotional exhaustion were associated with adverse performance (i.e., low client satisfaction and low productivity, and high personnel costs). Contrary to our expectations, neither average levels of employee satisfaction nor emotional exhaustion were related to organizational efficiency. One explanation for this lack of effects is that efficiency is a composite concept that is derived from many other concepts (e.g., productivity, personnel costs, overhead, costs of housing, and so forth), many of which were not included in the present study. For some of these concepts (i.e., productivity and personnel costs) the expected findings were obtained. This suggests that whereas these two performance concepts did show the expected results, these two factors are counter-balanced by one or more unknown factors or processes.

Study limitations

The most important limitations of this study are the following. First, the study employed a cross-sectional design, as all concepts studied here pertained to largely the same time span. The client and employee surveys were conducted simultaneously (summer to autumn of 2004); the financial survey was conducted early in 2005, but was based on data referring to the whole of 2004. This implies that the causal direction of effects could not be established. Relevant to this issue, Schneider, Hanges, Smith, and Salvaggio (2003) reported in a longitudinal study that employee satisfaction and organizational performance were reciprocally related, whereas Koys (2001) reported that employee attitudes longitudinally affected organizational effectiveness, rather than the reverse. Given the scarcity and ambiguity of longitudinal evidence in this area, it is difficult to argue for one type of causal relationship in particular, underlining the need for additional longitudinal research.

A second limitation concerns the multi-level problem (Bliese & Jex, 1999). Our organizational-level expectations were largely derived from individual-level findings. For example, whereas in this study organizations with satisfied employees were evaluated more positively by their clients, it does not follow that our *interpretation* of this association ("satisfied employees feel more committed/obliged to their organization, have a higher motivation, and work therefore harder, which results in better individual performance and higher client satisfaction") is correct, as this interpretation is based on notions and findings that pertain to a lower aggregation level. It is difficult to circumvent this problem statistically. However, the individual-level relationships among work characteristics and employee well-being and performance have already been supported in previous research (e.g., De Lange et al., 2003; Judge et al., 2001). This implies that only the step from individual to organizational performance presents difficulties. In the introduction to the present study, we discussed several mechanisms that might influence the relationship between organizational and individual performance. Previous work supports the reasoning

that individual-level and organizational performance are related (among others, Koys, 2001), but the specific mechanisms accounting for these cross-level associations are largely unknown territory. This underlines the need for additional research into these mechanisms.

Finally, whereas the intraclass correlation coefficients calculated in this study showed that aggregation of individual-level scores was warranted (all values for ICC(1) were significant, and all values for ICC(2) exceeded .70, Klein & Kozlowski, 2000), especially the values for ICC(1) were low. Thus, the organizational level accounts for only a minor part of the individual-level variability on the study concepts. It is plausible that these low ICC(1) scores are due to the fact that we aggregated scores across a very large unit, consisting of several sub-units (departments). It is very possible that the average scores on the study concepts differ across the sub-units of an organization, and aggregation across these sub-units will neglect these differences: the scores of these sub-unit scores may even cancel each other out at the organizational level. The implications of this limitation for the present research is unclear. On the one hand, neglecting the between sub-unit variance may have led to a restriction of range in the organizational-level variables, suggesting that our findings present a conservative estimate of the true magnitude of the associations studied here. On the other hand, our analyses still revealed a substantial number of theoretically meaningful effects, suggesting that this problem did not severely affect our findings.

Study implications

In spite of these problems, the present findings extend current knowledge in at least four respects. Firstly, this study provides evidence that organizations in which employees experience high levels of emotional exhaustion (as a measure of employee job-specific well-being) perform sub-optimally, although the causal direction of this association is not clear. Note that previous research on the relationships between burnout/emotional exhaustion and performance has typically focused on individual-level performance, and in this sense the present study is among the first to suggest that the average level of employee burnout in the organization matters for organizational performance. The scarcity of research into the relationship between employee well-being and organizational performance also means that future research should (a) employ longitudinal designs to unravel the causal relationships among these concepts, and (b) examine the processes linking employee well-being and organizational performance.

Secondly, our two indicators of job-specific well-being retained different relationships with our performance measures. Whereas aggregated levels of employee satisfaction were positively related to client satisfaction (Heskett et al., 2002; Loveman, 1998) but negatively with organizational productivity, aggregated levels of emotional exhaustion related in a consistent way to our measures of organizational performance. This suggests that (a) different measures of employee well-being may retain different relationships with organizational performance, and (b) the aggregated level of emotional exhaustion is a more consistent predictor of organizational performance than job satisfaction. At a more general level, this suggests that being *unable* to perform (due to high levels of emotional exhaustion) is more important a predictor of performance than being *unwilling* to perform.

Thirdly, we distinguished among four performance indicators. Three of these (efficiency, personnel costs and productivity) primarily focus on the quantitative aspects of performance, i.e., organizational performance is good when the tasks are done quickly and cheaply. Interestingly, these types of performance would intuitively seem to be at odds with good performance on the fourth performance indicator, namely the clients' evaluations of the services delivered. These services include not only helping clients with washing themselves and taking care of household chores, but also the provision of informational and emotional support. As providing such support may be time-consuming (simply put, talking to people costs time), high scores on the three quantitatively-oriented performance measures could be negatively associated with the clients' evaluations of the service provided by the organizations. Relevant to this issue, the organizational-level correlations between client satisfaction and the three other performance indicators presented in Table 1 are low and not statistically significant. This suggests that organizations whose employees on average perform well in terms of providing emotional and informational support do not automatically obtain low scores on efficiency, productivity and personnel costs. Thus, it is possible for organizations to obtain high scores on both quantity- and quality-related performance measures.

Fourthly, from a practical perspective our findings underline the need for organizational policies in order to manage employee well-being. Previous research in the home care sector has shown that especially work-directed interventions (i.e., interventions deal with the structure of the tasks to be conducted) can be effective in reducing levels of emotional exhaustion and job demands in these organizations (Taris et al., 2004). Our study suggests that it is important to keep employees happy, both in terms of job satisfaction and levels of emotional exhaustion: previous findings suggest that this is especially a matter of creating the right levels of demands, control and support (cf. De Lange et al., 2003). Thus, our findings underline the importance of improving worker well-being; this is not only important for individual workers, but may also have positive consequences for organizations and their clients.

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